



Combinational logic circuits

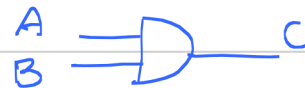
Logic gates are circuits that combine several logic variable inputs to produce a logic variable at the output.

AND Gate

AND gate performs logical multiplication. The output of the AND gate is 1 only if all inputs are 1.

All possible combinations may be listed in a table called truth table.

A	B	$C = A \cdot B$
0	0	0
0	1	0
1	0	0
1	1	1



circuit symbol for 2-input
AND gate

Properties of AND operation:

$$A \cdot A = A$$

$$A \cdot 1 = A$$

$$A \cdot 0 = 0$$

$$A \cdot B = B \cdot A$$

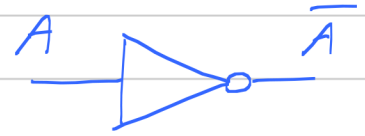
$$A \cdot (B \cdot C) = (A \cdot B) \cdot C = A \cdot B \cdot C$$

A	B	C	$A \cdot B \cdot C$
0	0	0	0
0	0	1	0
0	1	0	0
0	1	1	0
1	0	0	0
1	0	1	0
1	1	0	0
1	1	1	1



NOT gate (Logic inverter)

The NOT gate output is inverted input.



properties:

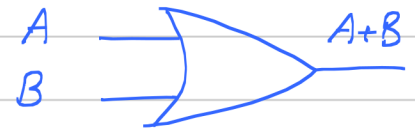
$$A \cdot \bar{A} = 0$$

$$\overline{\bar{A}} = A$$

A	$\bar{A}$
0	1
1	0

OR gate

The output of an OR gate is 0 only if all inputs are 0. A or B is written as $A+B$.



Properties:

$$(A+B) + C = A + (B+C) = A+B+C$$

$$A(B+C) = AB + AC$$

$$A + 0 = A$$

$$A + 1 = 1$$

$$A + \bar{A} = 1$$

$$A + A = A$$

A	B	$A+B$
0	0	0
0	1	1
1	0	1
1	1	1

NAND gate

NAND operation is the inversion of the result obtained from AND operation. The result is 0 only when all inputs are 1, otherwise it is 1.

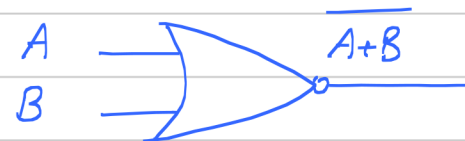
A	B	$\overline{AB}$
0	0	1
0	1	1
1	0	1
1	1	0



NOR gate

NOR operation is the inversion of the result obtained from OR operation. The result is 1 only when all inputs are 0, otherwise it is 0.

A	B	$\overline{A+B}$
0	0	1
0	1	0
1	0	0
1	1	0



XOR gate

XOR (Exclusive OR) operation produces 0 at the output only when both inputs are equal, otherwise it is 1.

A	B	$A \oplus B$
0	0	0
0	1	1
1	0	1
1	1	0



$$A \oplus B = A\overline{B} + \overline{A}B$$

Boolean Algebra

Manipulation of logic variables by AND, OR, NOT and other operations is different from ordinary algebra. In 1849 George Boole developed the mathematics of logic variables which is known as Boolean algebra which is based on logical thought and reasoning.

Boolean algebra is the algebra of two values 0 (or False) and 1 (or true).

Rules of Boolean algebra:

$$\overline{\overline{A}} = A$$

$$A \cdot A = A$$

$$A \cdot \overline{A} = 0$$

$$A \cdot 0 = 0$$

$$A \cdot 1 = A$$

$$A + \overline{A} = 1$$

$$A + 0 = A$$

$$A + 1 = 1$$

$$A + A = A$$

$$\text{commutative} \begin{cases} x \cdot y = y \cdot x \\ x + y = y + x \end{cases}$$

$$\text{Associative} \begin{cases} x \cdot (y \cdot z) = (x \cdot y) \cdot z \\ x + (y + z) = (x + y) + z \end{cases}$$

$$\text{Distributive} \begin{cases} x \cdot (y + z) = x \cdot y + x \cdot z \\ x + (y \cdot z) = (x + y) \cdot (x + z) \end{cases}$$

De Morgan's Laws

$$\overline{A + B + C} = \overline{A} \cdot \overline{B} \cdot \overline{C}$$

$$\overline{A \cdot B \cdot C} = \overline{A} + \overline{B} + \overline{C}$$

To prove that two logic statements are equal we need to show that they have identical truth table.

Ex. Prove $A(B+c) = AB + AC$

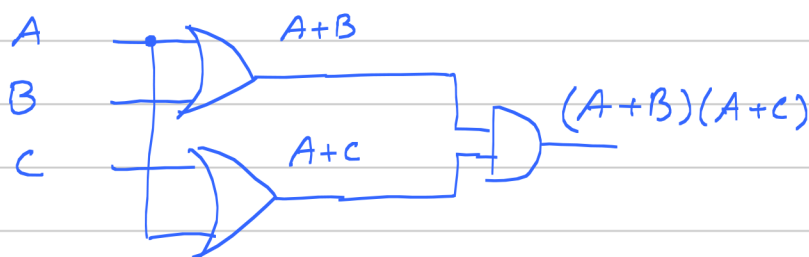
A	B	C	B+c	A(B+c)	AB	AC	AB+AC
0	0	0	0	0	0	0	0
0	0	1	1	0	0	0	0
0	1	0	1	0	0	0	0
0	1	1	1	0	0	0	0
1	0	0	0	0	0	0	0
1	0	1	1	1	0	1	1
1	1	0	1	1	1	0	1
1	1	1	1	1	1	1	1

identical

Implementation of Boolean Algebra

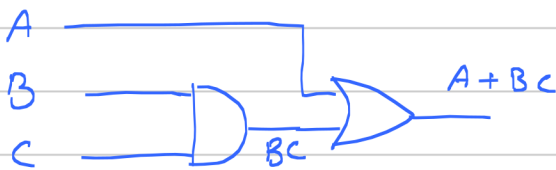
A Boolean expression can be implemented using a logic circuit with AND, OR, and NOT gates.

Ex. Make a logic circuit to implement $D = (A+B)(A+c)$



we can simplify the expression before implementing it.

$$\begin{aligned} D &= (A+B)(A+C) = AA + AC + BA + BC \\ &= A + AC + AB + BC \\ &= A(1 + C + B) + BC \\ &= A(1) + BC \\ &= A + BC \end{aligned}$$



Ex. use De Morgan's law to find the alternative expressions for

$$D = \overline{(\overline{A} + \overline{B}) \cdot (B + \overline{C})}, \quad E = \overline{[F(G + \overline{H}) + F\overline{G}]}$$

$$\begin{aligned} D &= \overline{(\overline{A} + \overline{B}) \cdot (B + \overline{C})} \\ &= \overline{\overline{A} + \overline{B}} + \overline{B + \overline{C}} \\ &= \overline{\overline{A}} \cdot \overline{\overline{B}} + \overline{B} \cdot \overline{\overline{C}} \\ &= AB + \overline{B}C \end{aligned}$$

$$\begin{aligned} E &= \overline{F(G + \overline{H}) \cdot F\overline{G}} \\ &= \overline{F + (G + \overline{H})} \cdot \overline{F + \overline{G}} \\ &= \overline{F} + \overline{(G + \overline{H})} \cdot \overline{F + G} \\ &= \overline{F} + (\overline{G} \cdot \overline{H} \cdot G) \\ &= \overline{F} + 0 \\ &= \overline{F} \end{aligned}$$

Ex. A laptop design has two processors (P1 and P2) and four slots for memory chips (S1, S2, S3, S4).

When processor P1 is running, it uses the memory in slot S1.

When processor P2 is running, it uses the memory in slot S4.

When both processors are running simultaneously, they use memory in slots S2 and S3.

Design a logic circuit to connect the processors and the memory slots. If a processor is running it produces high output to activate the memory slots.

Truth table :

P_1	P_2	S_1	S_2	S_3	S_4
0	0	0	0	0	0
0	1	0	0	0	1
1	0	1	0	0	0
1	1	0	1	1	0

$$S_1 = P_1 \bar{P}_2, \quad S_2 = P_1 P_2, \quad S_3 = P_1 P_2, \quad S_4 = \bar{P}_1 P_2$$

